

Crosscheck — 2-Minute Demo Script

A self-improving research copilot on **cognee**: persistent memory + a hero feature that flags when sources contradict each other.

Setup (before recording)

- Ollama running with `llama3.1:8b` and `nomic-embed-text`.
- From the repo root:

```
set -a && source .env && set +a
source .venv/bin/activate
uvicorn crosscheck.api:app --port 8010
```

- Open **`http://localhost:8010`**. The graph viz is served in-app at `/graph`.

The FooDB demo pack is already ingested and **persists across restarts**, so the graph and the contradiction card load immediately — no waiting on camera.

The script (data already persisted — leads with the hero)

1. Open the app — pause on the graph (~15s)

"This is Crosscheck's live memory — 20 nodes, 36 edges. It persisted across a restart; nothing was re-ingested."

Point out the Story/Type layout: sources `srcA-2021` and `srcD-2024` both wire into the `foodb` and `metric` nodes — the contradiction is visible in the graph.

2. Contradictions (§3) — the hero (~40s)

The  card is already showing:

foodb — *srcA-2021 (2021-03-01): 50,000 requests per second*
vs srcD-2024 (2024-09-01): 10,000 requests per second
"...same metric and time frame, but different values."

"The hero: it caught two sources disagreeing on FooDB's throughput — who said what, and when. Claims are extracted straight from the source text (keeping the number + its origin), then a

structural pre-filter and an LLM judge confirm the pair can't both be true."

Optional: click the ... refresh to show it recomputing live.

3. Gaps to research (§4) — the self-improving loop (~30s)

Click **Suggest**. Grounded questions appear:

"It ranks the sparsely-connected nodes and asks what to research next — database performance, MIT vs copyleft licensing, throughput metrics."

4. Graph — live cognee memory (~20s)

Drag a node; switch the **Graph / Schema / Memory** tabs at the top.

"All of this is real cognee memory — storage, the knowledge graph, and cited retrieval, running fully offline on a local Llama."

Cautions for the recording

- **Skip Ask (§2) on camera.** `/ask` is flaky on a small local model (cognee's search-completion path uses a `str | None` field BAML can't map on llama3.1:8b). It works on a hosted model; the demo leads with Contradictions.
- **Gaps' 5th question** can drift off-topic (cognee over-extracted a spurious concept node whose neighbours are noise). Stop at 4 or scroll past it.
- Google Fonts / favicon 404 console warnings are cosmetic (offline font fallback).

Alternative: film the "empty → Ingest → graph grows" narrative

Stronger story, but ~2-3 min of live cognify on Ollama with some on-camera risk. Reset first (prune the store), then start from the in-app onboarding: Ingest the demo pack → watch the graph populate → Contradictions → Gaps.

Why the architecture (30s if asked)

cognee's knowledge graph can't be the source of truth for a quantitative contradiction on a weak local model: the extractor flattens "50,000 requests per second" into a generic node and merges entities across sources, so the conflicting numbers never survive into the graph. Crosscheck reads faithful (subject, predicate, object) claims from raw text instead — verbatim number + source id +

timestamp — and feeds those to the structural + judge pipeline. cognee still powers storage, graph visualization, and cited search.